

In-situ Progressive Evaluation of VR Immersion: A Comparison between Voice and Controller-based Questionnaires validated with EEG

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Figure 1: Overview of the experimental set-up comparing Controller-based (left) and Voice-based (right) in-VR questionnaires, with real-time EEG physiological monitoring (note the headband device) to assess VR UX.

ABSTRACT

In-situ VR surveys offer ecological validity compared to post-stimulus questionnaires. However, surveying during the ongoing task can cause interruption and bias measured variables. Thus, in-situ data collection is designed to be simple and quick - typically relying on controller-based Likert-scale responses to short 2D pop-up questions. This pilot study proposes using voice communication instead, validating its effectiveness in maintaining gradual immersion without disrupting user experience. Results indicate voice interaction enabled a more stable, stress-free immersive state compared to the controller-based method.

Index Terms: VR Evaluation, EEG, Multimodal Interaction, Voice User Interface, Presence, Pilot Study.

1 INTRODUCTION

In-situ VR surveys are essential for ecological accuracy but often disrupt user immersion due to visual distraction and the cognitive load of controller manipulation [1, 2]. To address this, we developed a non-visual voice-based survey system validated by real-time EEG monitoring. We hypothesize that voice interaction minimizes mental context switching, enabling better maintenance of immersion compared to traditional controller-based methods, as verified by physiological signals.

2 RELATED WORK

Brown and Cairns [3] defined immersion as a graded experience progressing through engagement, engrossment, and total immersion. To objectively measure these states, recent studies have utilized EEG; for instance, Katahira et al. [4] correlated Flow states with increased frontal theta and alpha rhythms, while Souza et al.

[5] validated EEG for detecting immersion levels. However, evaluating immersion in-situ remains challenging. Safikhani et al. [2] noted that in-VR questionnaire design significantly impacts user experience. Interruptions necessitate "Situation Awareness Recovery" [6] to regain immersion. This suggests that, unlike controller-based interactions, which may impose high cognitive load, natural modalities like voice could facilitate the maintenance of the immersive state by minimizing the recovery phase.

3 EXPERIMENT

We employed a 1 factor between-subjects design to isolate the impact of interaction modality on immersion during survey response collection (see Figure 1): (1) "Controller" (subjects responded to pop-up questions with answers selected by ray casting) and (2) "Voice" based interaction (system presented questions in voice, answered in voice, and recognized). A total of eight participants (4 Male and 4 Female) were recruited and assigned to one of the two groups. Each session lasted approximately 15 minutes with three survey prompts (time-triggered for Controller, operator-triggered for Voice). Survey items were adapted from the Immersion Experience Questionnaire [7] (see supplementary materials).

Participants & EEG Processing: Eight participants (4M/4F) played *Angry Birds VR: Isle of Pigs* wearing the Looxid Link 2.0 EEG headband. Signals were recorded from Fp1 and Fp2 channels, preprocessed with a 60Hz notch and 1~45Hz bandpass filter. Power spectral density was analyzed across delta (1~4Hz), theta (4~8Hz), alpha (8~13Hz), beta (13~30Hz), and gamma (30~45Hz) bands. Metrics (Stress, Emotional Stability, Flow) were processed by removing artifacts, resampling to 1s intervals, and Min-Max normalization (0~1).

Immersion Classification: Immersion states (Overload, Boredom, Flow, Engrossment, Engagement) were classified by applying rule-based thresholds to the normalized EEG metrics. This operationalizes the Brown and Cairns' theoretical model [3] that frontal EEG band ratios can effectively discern immersion levels in VR [5].

Statistical Analysis: Group comparisons used Welch's t-test with Cohen's d. Immersion slope was derived via OLS linear regression of survey scores over time to quantify stability (slope ≈ 0) vs. recovery efforts (positive slope). Post-hoc power analysis in-

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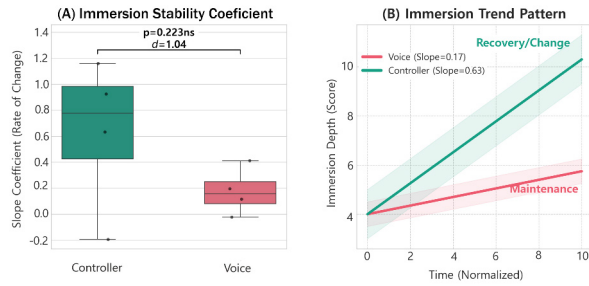


Figure 2: **Immersion Dynamics.** (A) **Immersion Stability Coefficient:** Controller shows higher values (instability) vs. Voice near zero (stability). (B) Trend patterns: “Recovery” vs. “Maintenance.”

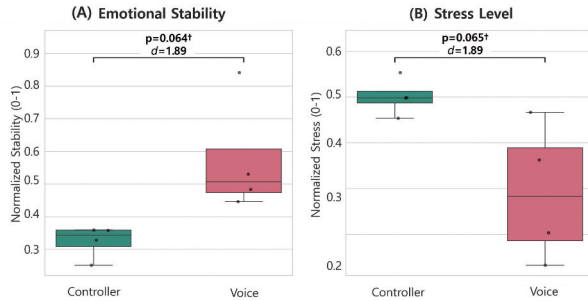


Figure 3: **Physiological Verification.** The Voice interaction demonstrates superior (A) Emotional Stability and marginally lower (B) Stress Level with a large effect size ($d = 1.89$), confirming that it induces a state of “Relaxed Immersion.”

icated 72% power for detecting large effects ($d \geq 1.0$) with $n = 4$ per group.

4 PRELIMINARY RESULTS

4.1 Immersion Dynamics: Recovery vs. Maintenance

To investigate how each interaction modality affects the continuity of the user’s experience, we analyzed the temporal pattern of immersion using the linear regression slope of in-situ focus probe scores (1–10 scale). The Controller group’s slope coefficients were notably higher ($M = 0.63$), indicating a “Recovery” pattern (see Figure 2 (A)). In contrast, the Voice group showed near-zero slope coefficients ($M = 0.17$) with a large effect size ($d = 1.04$), confirming a “Maintenance” pattern. Figure 2(B) illustrates these contrasting trend patterns. Consequently, these results strongly demonstrate that voice interaction is superior in sustaining immersion compared to the controller method. Further analysis (dynamic range, volatility) confirmed Controller group exhibited notably higher fluctuations across metrics.

4.2 Physiological Evidence: Relaxed Immersion

To validate these patterns physiologically, we analyzed EEG data (see Figure 3). As for the Comfort (stress & emotion), the Voice group demonstrated marginally lower stress ($p = .065$) and higher emotional stability ($p = .064$). Despite the limited sample size preventing statistical significance at $\alpha = .05$, the very large effect sizes ($d = 1.89$) suggest a substantial practical difference in user comfort between modalities. For the Depth (Flow), there was no significant difference in the absolute depth of flow ($p = .60$). This indicates that the advantage of Voice UI lies not in “deeper” immersion, but in “more comfortable” immersion.

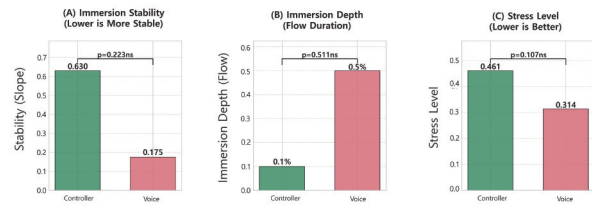


Figure 4: **Wholistic Comparison.** While Immersion Depth was comparable, the Voice interface demonstrated superior Stability (lower slope) and Comfort (lower stress) - that is, Voice UI facilitates “Relaxed & Sustained Immersion.”

4.3 Stability and Comfort

A wholistic comparison, using the behavioral and physiological data together) of the two interfaces (see Figure 4). While the Depth of immersion was comparable due to task characteristics, the Voice interface demonstrated superior Stability (indicated by a lower slope) and Comfort (indicated by lower stress). Consequently, we conclude that the Voice UI is an effective tool for facilitating “Relaxed and Sustained Immersion” in VR.

5 DISCUSSION AND CONCLUSION

We proposed a voice-based in-situ VR UX method demonstrating ecological validity for evaluating the gradual build-up of presence/immersion and VR UX. In particular, through the associated EEG data, we confirmed the lesser extent of disruption (less variability) despite the periodic data collections. This, in turn, would allow more detailed observation of the user experience in terms of engagement, engrossment, and immersion. Our results are still preliminary (e.g., low sample size) and also assumes the validity of the EEG analysis and rule-based classification. We acknowledge the inherent inequality in interaction difficulty; however, this contrast precisely highlights the cognitive cost of standard controller methods compared to the voice modality. For future work, we plan to try increasing the user’s burden and have the user think aloud of the VR UX and be analyzed by a multimodal AI agent (e.g., body gestures and facial expression). This would make the VR UX evaluation more multi-faceted and comprehensive, and thus reliable.

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